

Level: Bachelor	Semester: Spring	Year : 2024
Programme: BE		Full Marks:100
Course: System Programming		Pass Marks: 45
		Time : 3hrs.

Attempt all the questions.

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| 1. | a) | Explain the two important hypothetical architectures with their important features. | 10 |
| | b) | Differentiate RISC and CISC architectures | 5 |
| 2. | a) | What do you understand by the assembler design options? Explain with examples if needed. | 8 |
| | b) | Explain the features of program relocation as one of the machine dependent assemblers features. | 7 |
| 3. | a) | Consider the following assembly language program | 10 |

Line	Symbol	Opcode	Exp
10	SUM	START	0
20		LDX,	#0
30		LDA	#0
40		+LDB	#TABLE 2
50		BASE	TABLE 2
60	LOOP	ADD	TABLE,X
70		ADD	TABLE2,X
80		TIX	COUNT
90		JLT	LOOP
100		+STA	TOTAL
110		RSUB	
120	COUNT	RESW	1
130	TABLE	RESW	2000
140	TABLE 2	RESW	2000
150	TOTAL	RESW	1
160		END	

Mnemonic	Opcode
LDX	04
LDA	00
LDB	68
STA	0C
ADD	18
TIX	2C
JLT	38
RSUB	4C

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|----|---|---|----|
| | i. | Show all data structures | |
| | ii. | Create an object code column with object code | |
| | iii. | Create an object file | |
| b) | What is Literal? Explain its handling during pass 1 and pass 2. | | 5 |
| 4. | Consider the following assembly language program | | 15 |

Line	Symbol	Opcode	Exp
10	STRCPY	START	2505
20	FIRST	LDX	ZERO
30	MOVECH	LDCH	STR1,X
40		STCH	STR2,X
50		TIX	FIFTEEN
60		JLT	MOVECH
70	STR1	BYTE	C'EOF'
80	STR2	RESB	15
90	ZERO	WORD	0
100	FIFTEEN	WORD	15
110		END	FIRST

Mnemonic	Opcode
LDCH	50
LDX	04
STCH	54
JLT	38
TIX	2C

- i. Fill column for location counter
- ii. Create object code column with object codes
- iii. Show all data structures
- iv. Create Object code file
- v. Load the program in memory

5. a) How unique labels are generated in Macro Processor? Explain with example. 5
- b) What is Recursive Macro Definition? How One Pass Macro Processor Handles it? Explain with example. 5
- c) Consider the macro definition given below and show macro expansion for the macro call statement "PRINT 69, F8". Show all data structures used by macro processor clearly. 5

PRINT	MACRO	&Ch, &Od
\$LOOP	TD	&Od
	JEQ	\$LOOP
	LDCH	#&Ch
	WD	&Od
	MEND	

6. a) What is object oriented programming? Explain the object diagram for assembler. 7
- b) Explain the scenario of the relocating loader for two machine architectures namely SIC and SIC/XE. Please refer to question number 3(a) and 4 for the references. 8
7. Write short notes on: (Any two) 2×5
- a) Macro Development Process
 - b) Polymorphism
 - c) Conditional macro expansion

(SP-2024 SP)